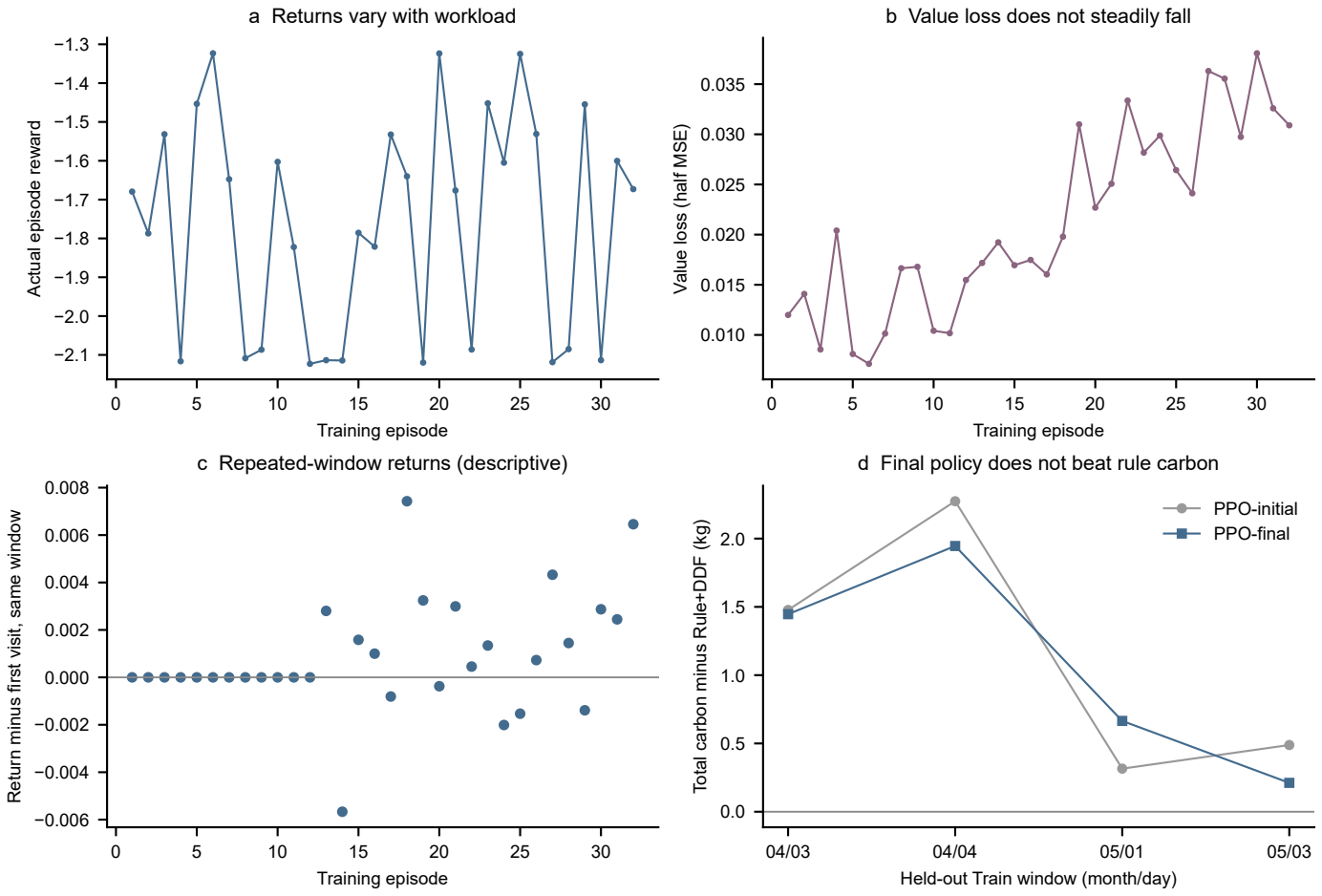
a Returns vary with workload
Actual episode reward
−1.3
−1.4
−1.5
−1.6
−1.7
−1.8
−1.9
−2.0
−2.1
0
5
10
15
20
25
30
Training episode
b Value loss does not steadily fall
Value loss (half MSE)
0.035
0.030
0.025
0.020
0.015
0.010
0
5
10
15
20
25
30
Training episode
c Repeated-window returns (descriptive)
Return minus first visit, same window
0.008
0.006
0.004
0.002
0.000
−0.002
−0.004
−0.006
0
5
10
15
20
25
30
Training episode
d Final policy does not beat rule carbon
PPO-initial
PPO-final
Total carbon minus Rule+DDF (kg)
2.0
1.5
1.0
0.5
0.0
04/03
04/04
05/01
05/03
Held-out Train window (month/day)